RGB → Point-Cloud generation: evaluation report

Checkpoint `outputs/4m_distill_rgb2pc/best_model.pth` (epoch 500) · teacher & student init `4m_run_v3` @ epoch 2400 · conditioned on **RGB only** (depth / PC / spline fully masked)

Result: the model memorises plants, it does not generalise to new ones

dataset	N	mean CD	median	best	worst
old10k / train — trained on these	150	0.00029	0.00029	0.00021	0.00040
old10k / val — *same plants, new camera views*	100	0.00027	0.00027	0.00022	0.00042
Sorghum_15K / test — unseen plants (OOD)	**150**	**0.04705**	**0.04280**	**0.01356**	**0.15145**

Full OOD sweep over all 22,500 test folders: mean **0.04987**, median 0.04375 (`per_plant_chamfer_15Ktest_ep500.json`).

> **The old 10k validation split is leaked.** It contains 100 folders covering only **10 distinct plants, and all 10 also appear in training** — the split was made per camera view, not per plant. Every "validation" sample is a training plant from a new angle, which is why train and val scores are identical (0.00029 vs 0.00027) instead of val being worse. The 0.0003 figure this run optimised against therefore never measured generalisation.

Verification

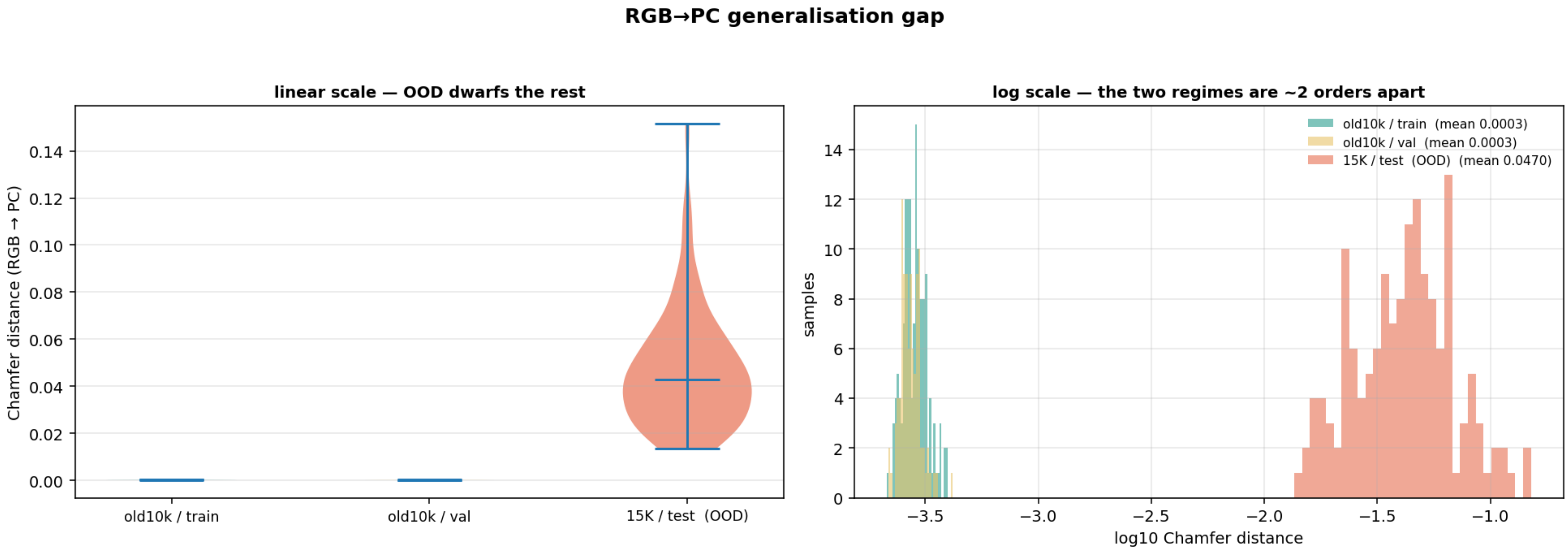
check	result
old10k train ∩ val (plants)	10 / 10 — fully leaked
Sorghum_15K train ∩ test (plants)	0 — leak-free
shared plant *IDs* old10k ↔ 15K test	137 (numbering reuse)
… of which identical geometry (spline md5)	0 / 60 sampled — genuinely unseen
render statistics old10k vs 15K	near-identical — no domain shift
Pearson r (Chamfer, phenotype extremeness)	0.017 — not tail extrapolation

Consequences

OOD Chamfer **rose** monotonically over training (0.0381 @ ep65 → 0.0453 @ ep240 → 0.0499 @ ep500) while in-domain fell ~10×: the run was memorising harder, not learning better. `best_model.pth` is selected on the leaked val loss, so it is the *worst* OOD checkpoint of the run. The two datasets are statistically indistinguishable at the pixel and point level, and error does not track phenotype extremeness — so this is neither a render-domain shift nor a tail-extrapolation failure. It is a plain failure to generalise across plant identity, hidden until now by the leaked split.

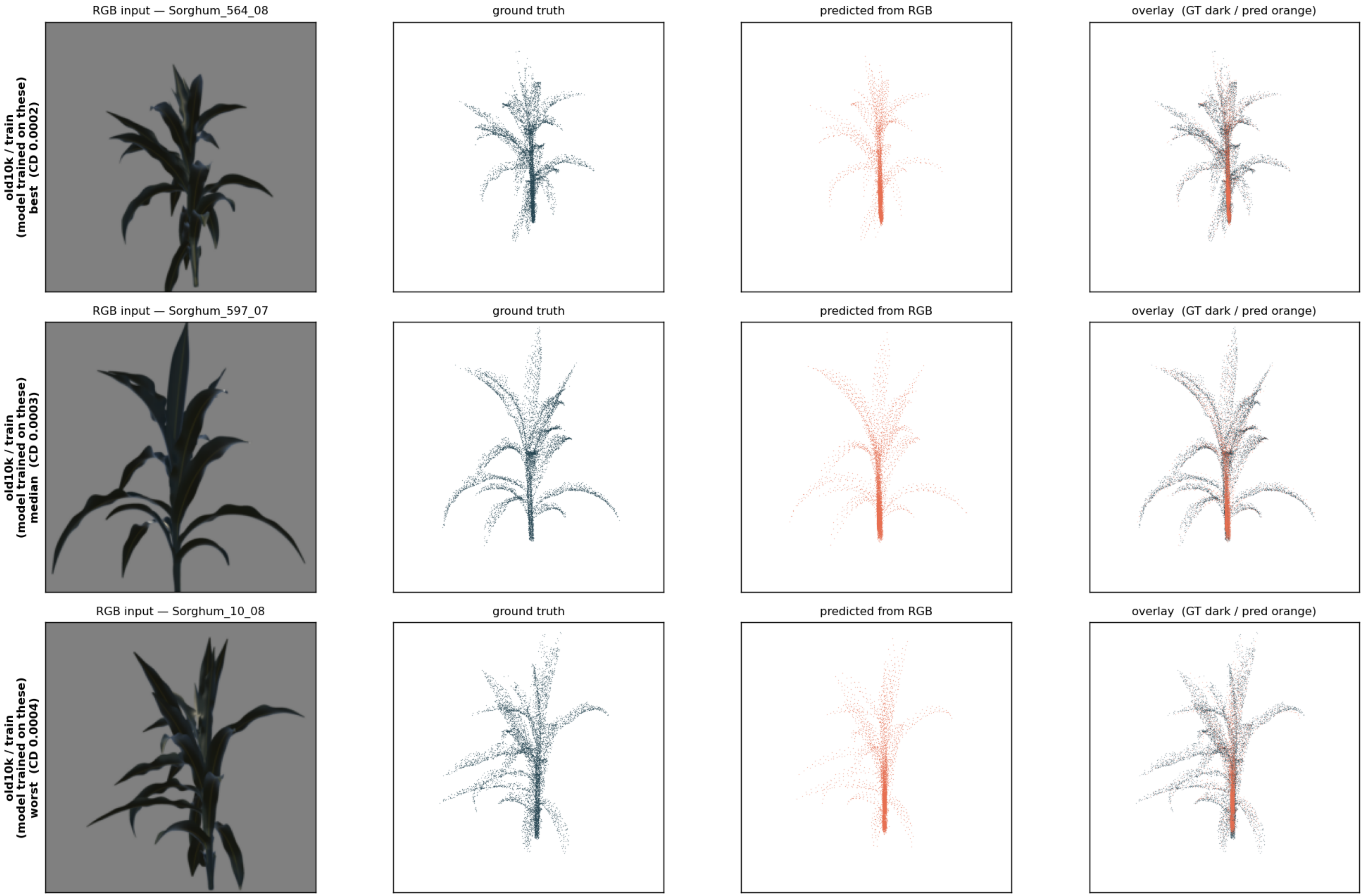
Next: redo the distillation on the 15K train split (leak-free, split by plant), using the in-flight 15k pretrain (job 11493779) as teacher, and select on the 15K val split.

Chamfer distributions — the two regimes do not overlap



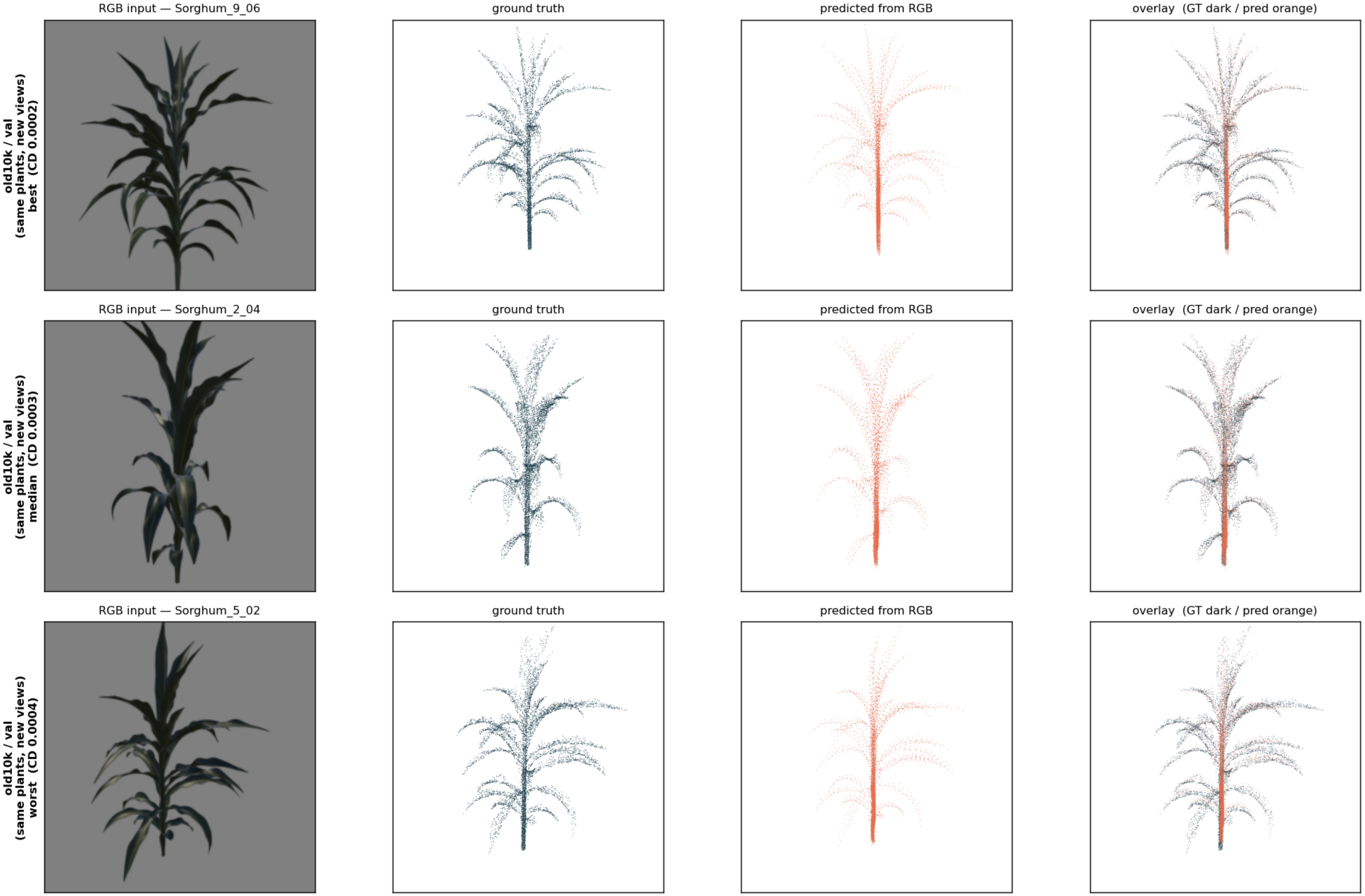
Equal N per dataset. On the log axis the in-domain and OOD populations sit ~2 orders of magnitude apart with no overlap: every unseen plant is reconstructed worse than every seen one.

old10k / train (model trained on these) — best / median / worst (epoch 500)



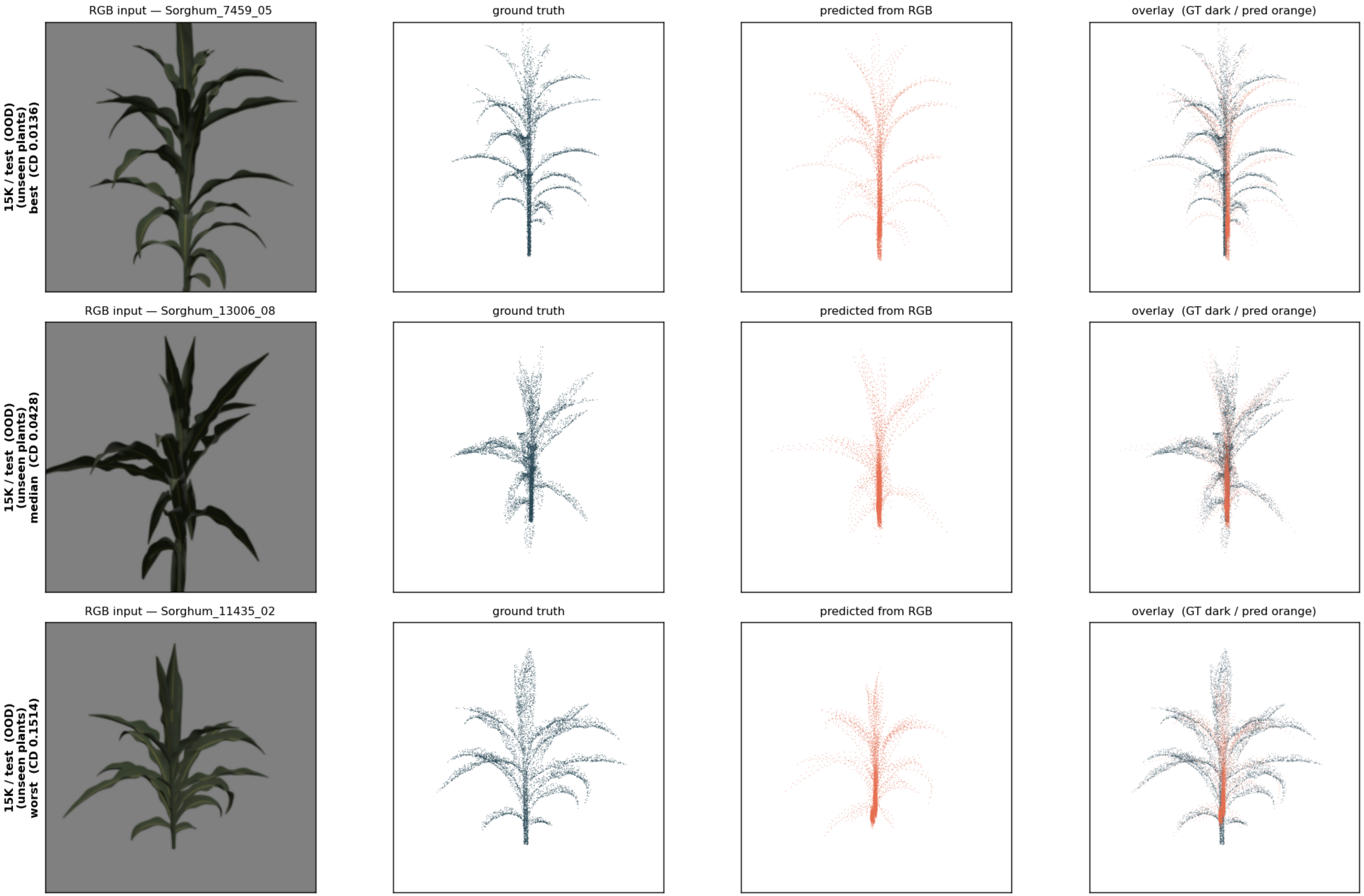
Trained on these plants. The predicted cloud (orange) lands on the ground truth (dark) almost everywhere, including leaf tips.

old10k / val (same plants, new views) — best / median / worst (epoch 500)



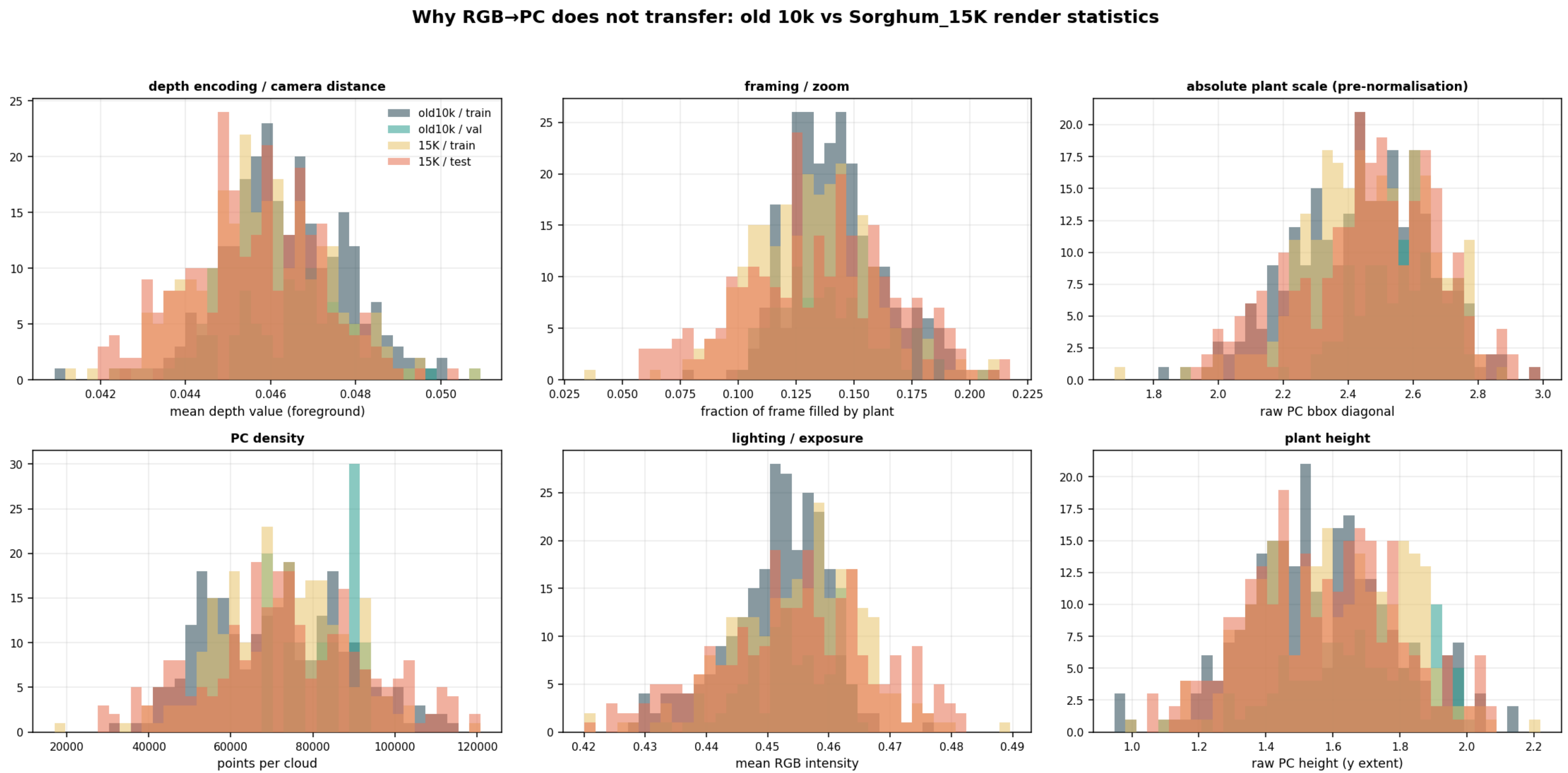
Same plants as training, new camera views — the leaked "validation" set. Indistinguishable from training performance, which is the tell.

15K / test (OOD) (unseen plants) — best / median / worst (epoch 500)



Unseen plants. The prediction collapses to a narrow, generic stem-and-fan shape: plant-like, but not THIS plant. Even the best OOD sample is ~32x worse than the worst in-domain one.

Control: the two datasets are the same render domain



Depth encoding, framing, raw point-cloud scale, point density and exposure all coincide across old10k and Sorghum_15K — ruling out a render-domain shift as the cause of the OOD collapse.